BUILD-EMS

Behind-the-Meter Control for Rooftop Solar, Wind, Storage & EV — Multi-Tenant Ready



Deploying EV charge/discharge with rooftop solar PV and wind on an existing building? **BUILD-EMS** is the open-source behind-the-meter controller that integrates **bidirectional DC/AC inverters**, multi-tenant "plug-in solar", and EV infrastructure into one balanced building minigrid — enforcing multi-zone energy policy while protecting the mains.

What Sets BUILD-EMS Apart:

- **Bidirectional & Inverter-Ready** : coordinates DC/AC bidirectional inverters and converters for solar PV, wind and battery, with EV charge and discharge (V2X).
- **Multi-Tenant "Plug-in Solar"** : partitions shared rooftop generation and storage across tenants with per-zone, time-of-day energy policies.
- **Mains-Protective** : balances the building mains and responds to grid-peak DERMS events without exceeding the service limit.
- **Open-Source Hardware** : released under CERN-OHL-S v2 — schematics, board files and firmware published on GitHub (meshEMS / X-EMS).
- **Field-Serviceable** : built from open, off-the-shelf commodity modules, snap-in mounted for fast in-country assembly and repair.

BUILD-EMS turns a building rooftop into a coordinated, multi-tenant minigrid — maximizing solar, wind and EV flexibility while keeping the mains within limits.

Intelligent

- Multi-zone, time-of-day energy policy
- Mains balancing & DERMS peak response
- EV charge / discharge (V2X) scheduling

Interoperable

- Bidirectional DC/AC inverters (PV / wind / battery)
- SunSpec DER + Modbus / CAN
- Multitenant sub-metering & EVSE

Resilient

- Local policy enforcement, cloud-optional
- Per-tenant energy allocation
- Open meshEMS platform

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Product Specifications

PRODUCT	
Product name	BUILD-EMS (10Power X-EMS, supplied by NESL)
Model / PCB	NESL EMS Controller PCB 865B (shared X-EMS platform)
Firmware	meshEMS / OpenAMI metering (open source, ~50k LOC middleware)
License & source	Hardware: CERN-OHL-S v2 | Repository: GitHub, meshEMS / X-EMS
Serviceability	Open, off-the-shelf commodity modules | snap-in mounted for fast in-country assembly and repair
COMPUTE	
Processor	Dual-core 32-bit MCU @ 240 MHz | WiFi + Bluetooth LE
Memory	16 MB Flash + 8 MB PSRAM
ENERGY MANAGEMENT	
Supported DER	Solar PV | wind | battery, via bidirectional DC/AC inverters / converters
EV infrastructure	EVSE charge & discharge (V2X) | per-tenant charging needs
Energy policy	Multi-zone time-of-day policies | per-channel kW / kWh limits | multitenant allocation | DERMS response
ASSET COMMUNICATION	
Serial buses	2× RS-485 Modbus: EV-charger (EVC) + sensor/DTM ports | 1× CAN 2.0
On-board metering	SPI poly-phase metering, up to 6 CT channels | bidirectional (import/export)
Local I/O	2× onboard SSR outputs | door-switch input | 2 analog buttons
SUPPORTED MODULES & SENSORS (out of the box)	
DER & inverters	SunSpec models 1 / 11 / 213 (bidirectional inverters & meters)
Energy meters / relay	Single-phase Modbus meters (bidirectional kWh) | 8-ch I^2C SSR bank
Sensors / networking	Temp / humidity sensor | Ethernet | WiFi / BLE
POWER & ENCLOSURE	
Operating voltage	5 V DC — USB-C or screw terminals | or 4.5 – 20 V DC | or 8 – 12 V AC (SELV)
Power / dimensions	85 × 55 mm | or Raspberry Pi HAT form factor
STANDARDS & POSITIONING	
Grid position	Behind-the-meter (BTM) | building / commercial rooftop minigrid
Capabilities	EV V2X | multitenant solar / wind / storage | mains balancing & DERMS

Customizable with added hardware modules and open or licensed code stacks. BUILD-EMS is an open-source X-EMS product supplied by New Energy Solutions Lab for 10Power.